

Respiratory emergency visits in Chile

A defensible five-year case

Selected W1 line · DEIS public emergency network · 2020-2024

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Executive summary

This report asks how the volume and share of **recorded respiratory emergency visits** changed by week and care setting in the Chilean public network from 2020 through 2024, and what the history can contribute to a proposed Winter Campaign monitoring decision. The unit is a consultation count reported by a facility, not a person, risk estimate or measure of beds and staffing. The analysis uses every date in the five local DEIS Urgencias archives for cause IDs 1 and 2, a 2,169,824-row uncleaned reduced input. [1, 6]

The respiratory share of all recorded emergency visits was 10.75% in 2020, 9.97% in 2021, and 24.69%-28.95% in 2022-2024. Complete-week respiratory volume peaked in May in both 2023 and 2024 (189,297 and 203,353 visits). The share pattern persists in a panel of 581 facilities present all five years. These observations support earlier monitoring review, but they do not establish why the pattern changed or prove that capacity was insufficient.

C1 corrects a W1 reproducibility gap. W1 had already reported five-year aggregates, yet its submitted reduced emergency input covered only selected peak weeks in 2023-2024. The expanded C1 case includes all dates in 2020-2024 for the two KPI codes, removes two identical 2023 duplicate rows from derived tables, and groups weeks from actual dates. W1 received 100/100; the wider-data suggestion was internal team feedback, and no specific instructor comments were reported.

1. Selected problem, prior alternatives and policy context

Respiratory illness can create changing demand across public emergency settings. A plausible proposed user is winter planning staff in the public care network. The bounded decision is when to intensify review of weekly demand by care setting before or during the Winter Campaign. Actual changes to shifts or beds require current capacity and clinical data that this case lacks. MINSAL's National Health Strategy discusses acute respiratory illness, and its Winter Campaign reporting shows the operational relevance of monitoring emergency consultations. Our respiratory-visit share is **not** a numerical target for morbidity or mortality in that strategy. [2, 3]

W1 option	Evidence and fit	C1 decision
A · Respiratory emergency visits	Weekly facility/cause counts in 2020-2024, with a direct historical demand measure; source reporting and pandemic years require caution.	Selected. High-frequency activity series can support bounded timing and monitoring questions.
B · Child nutrition in REM-P2	June/December cuts for children under six, with coverage and codebook issues documented in W1.	Retained as the W1 alternative. Its sparse cuts were a weaker fit for a short-horizon analytical continuation.

The documented W1 selection received a full-mark result. There were no specific instructor comments to accept or reject. A teammate's subsequent internal review identified that the packaged W1 reduced emergency input did not support independent reconstruction of all five years. C1 responds by supplying the full-date selected input, its preparation scripts and executed notebook. The analytical question remains focused on recorded demand; geographic comparison is deferred because comparable 2020-2022 geographic fields and a validated historical establishment join are not included.

2. Seven steps from Class 03

Step	Case-specific reasoning
1. Decision	Proposed user: public-network winter planning staff. Decision: when to review preparedness and intensify monitoring by setting; staffing and bed changes need additional evidence.
2. Success criteria	Desired service improvement is earlier, better-informed preparation. C1 can verify a consistent historical weekly series, its peaks, shares and coverage; it cannot measure improved outcomes.
3. Business questions	Main: how did weekly volume and share change in 2020-2024? Supporting: when were complete-week peaks; how did age and care-setting mix change; does the broad share pattern persist in common facilities?
4. Business process	Facilities record consultations; DEIS publishes grouped daily facility/cause activity. The files omit patient trajectories, wait times, staff, beds, outcomes and unrecorded demand.
5. Entities, events and states	Facility is an entity; a consultation is the counted event; each source row is an aggregate facility/date/cause state. Visits are not unique people.
6. Data representation	Candidate source key: year + facility ID + date + cause ID. Counts are Total and five age bands; facility type is a care-location category. Annual files are harmonized without an external join.
7. Measurement	Weekly respiratory visits = sum of Total for ID 2. Respiratory share (%) = $100 \times \text{ID 2} / \text{ID 1}$ for the same period and scope. Age/type shares use all respiratory visits as denominator. Exclude incomplete weeks for peak comparisons.

3. Data, definitions and preparation

The exact source versions are five DEIS AtencionesUrgenciaYYYY ZIP files, retrieved 24 September 2026. `data/source\_manifest.csv` records each URL, byte size, SHA-256 and retrieval time; the official dictionary is in `data/reference/`. The selected input retains every raw ID 1/2 row from every date in local 2020-2024 archives. ID 1 is all recorded emergency visits; ID 2 is the respiratory subset. `Total` equals the sum of five age-band counts for every selected row. The data are grouped records from the public network, not a patient-level file. [1, 6]

Preparation order: (1) filter only IDs 1/2 from annual originals while retaining selected source rows unchanged in the reduced extract; (2) profile keys, dates, measures, recorded zeros and facility coverage; (3) remove one extra copy of each of two identical 2023 facility/date/cause rows; (4) parse dates and derive Sunday `week\_start`; (5) aggregate to facility-week-cause and national-week tables; (6) calculate shares only after numerator and denominator are aligned. Source originals remain unchanged. The 2023 duplicate removal subtracts 13 from all visits and zero from respiratory visits. Every derived annual count reconciles to the weekly tables.

Measure	Numerator and denominator	Unit / interpretation
Weekly respiratory volume	Sum of Total where IdCausa = 2 within one Sunday-Saturday week.	Recorded visits; incomplete first/last weeks are flagged.
Respiratory share	$100 \times \text{Total}(\text{ID 2}) / \text{Total}(\text{ID 1})$, same date scope and included facilities.	Percentage of recorded emergency activity, not incidence or risk.
Age composition	$100 \times \text{respiratory age-band count} / \text{all respiratory visits}$ in same scope.	Distribution of visits; age cells reconcile to Total.
Care-setting mix	$100 \times \text{respiratory visits at a facility type} / \text{all respiratory visits}$.	Location of care, not home residence, occupancy or quality.

4. Profiling, preparation checks and the Five Cs

The 2,169,824 selected raw rows represent 1,084,911 distinct facility-dates after duplicate removal. Every such facility-date has both IDs 1 and 2, and respiratory counts never exceed all-visit counts. Key, measure and facility-type fields have no nulls; dates parse within their source years; weeks are in 1-53; no negative cells or age-sum mismatches were found. This is an observed check, not proof that every real consultation was reported. Recorded zeros remain: ID 2 has 38,434 zero-count rows in 2020 and 14,800 in 2024. Two identical 2023 extra rows, one per cause ID, were removed only from processed outputs.

C1 lens	Check and consequence
Clean	Check keys, dates, numeric validity, age sums, zeros and exact duplicates. Remove identical 2023 extras; annual ID 1 falls by 13, ID 2 by 0.
Consistent	Use the same cause and age definitions across selected annual archives. Annual `semana` labels can map to disjoint boundary weeks (2022 label 52, 2023/2024 label 1); derive weeks from dates instead.
Conformed	Harmonize the shared selected fields and IDs across all five files. No geographic facility-register join is used: it would not answer the national question and would require historical key/cardinality validation.
Current	Historical 2020-2024 versions suit retrospective seasonality. A same-week 2026 decision would need a current feed, version/freshness flags and operational capacity data.
Comprehensive	All local years/dates for IDs 1/2 are included. The data omit unreported care, private activity outside this public-network scope, unique patients, residence, population denominators, staffing, beds and outcomes.

Year	Selected source rows	ID 2 zero rows	Duplicate extras	Facilities
2020	389,082	38,434	0	607
2021	440,812	46,616	0	635
2022	446,316	21,424	0	638
2023	444,954	15,157	2	642
2024	448,660	14,800	0	640

Table 1. Source-row profile before duplicate removal. Zero-count rows are recorded values, not missing observations. The two 2023 extras are identical source copies.

Facility reporting coverage is uneven. There are 607 reporting facility IDs in 2020 and 640 in 2024; 41 report fewer than 100 days in 2020 versus 15 in 2024. This cautions against reading absolute count differences as a pure change in underlying illness. The 581 IDs present all five years account for 98.05% of all visits in 2020 and 95.89% in 2024; their share pattern is close to the full sample.

5. Exploratory results and interpretation

Year	All visits	Respiratory visits	Respiratory share
2020	11,382,496	1,223,447	10.75%
2021	13,851,594	1,381,568	9.97%
2022	19,491,232	4,812,171	24.69%
2023	19,194,227	5,557,541	28.95%
2024	19,436,048	5,541,628	28.51%

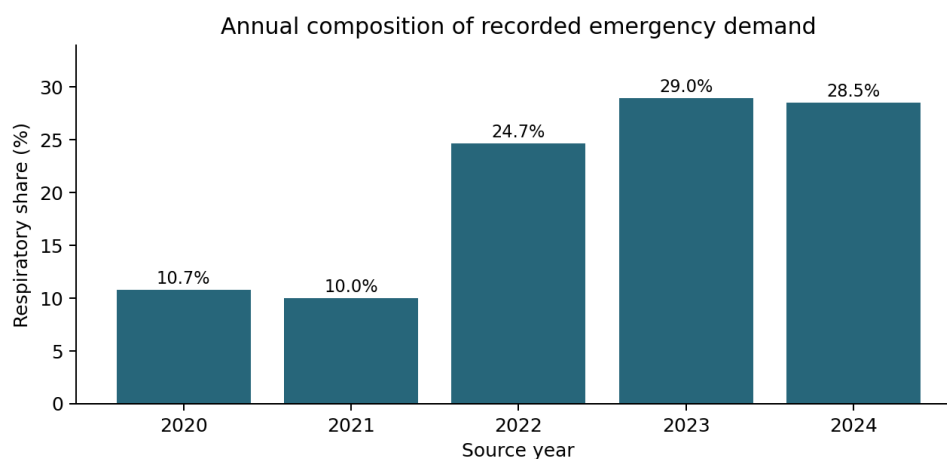


Figure 1. Respiratory visits as a percentage of all recorded emergency visits by source year. 2020-2021 should not be treated as an ordinary seasonal baseline without considering the pandemic-era regime.

Observed: the respiratory share is around 10% in 2020-2021 and 24.69%-28.95% in 2022-2024.

Interpretation: the mix of recorded emergency activity changed substantially; comparing years requires care because reporting facilities, care seeking and pandemic-era conditions may differ. **Not established:** a change in individual disease risk, clinical severity, policy effect or service capacity.

Year	Week start	Week end	Respiratory visits	Share
2020	15 Mar	21 Mar 2020	103,314	38.04%
2021	14 Nov	20 Nov 2021	47,697	14.61%
2022	30 Oct	05 Nov 2022	161,522	36.00%
2023	14 May	20 May 2023	189,297	41.66%
2024	12 May	18 May 2024	203,353	45.56%

Table 2. Largest respiratory-volume week in each reporting year among 260 complete Sunday-Saturday weeks. The first and last generated weeks have only four and three source days and are excluded from this comparison.

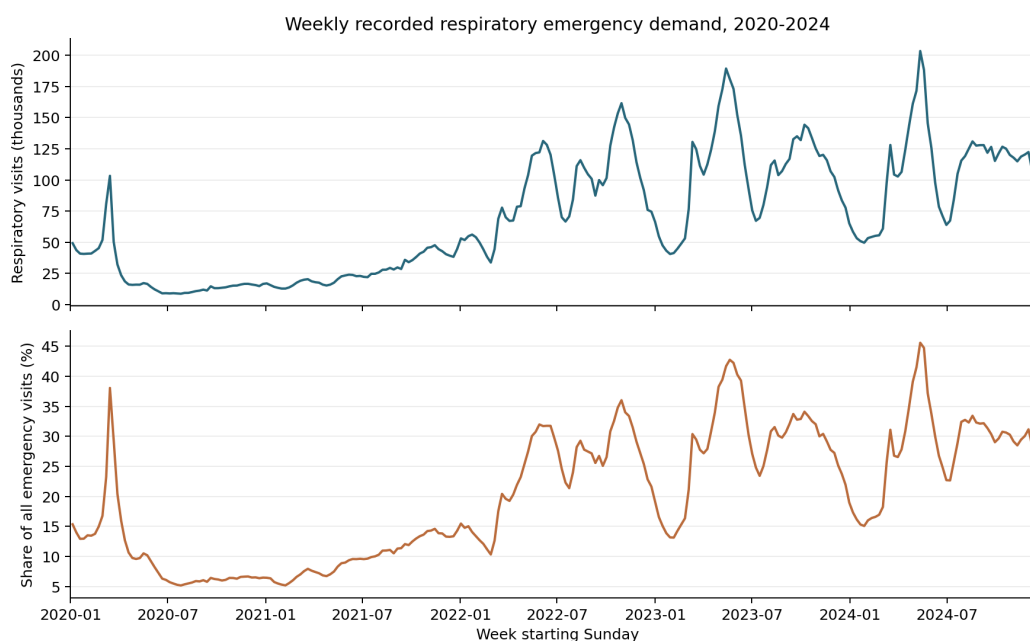


Figure 2. Weekly recorded respiratory volume and share from the selected five-year case. The shape is descriptive; a seasonal forecast is outside C1.

The 2023 and 2024 complete-week peaks occurred in May, with 189,297 (41.66%) and 203,353 (45.56%) respiratory visits respectively. MINSAL's 24 May 2024 campaign report records a 45.4% week-20 respiratory share. That is close but not numerically identical to 45.56% from this downloaded DEIS archive; differing version, scope or definitions have not been resolved, so the external report corroborates timing rather than an exact number. [3, 4]

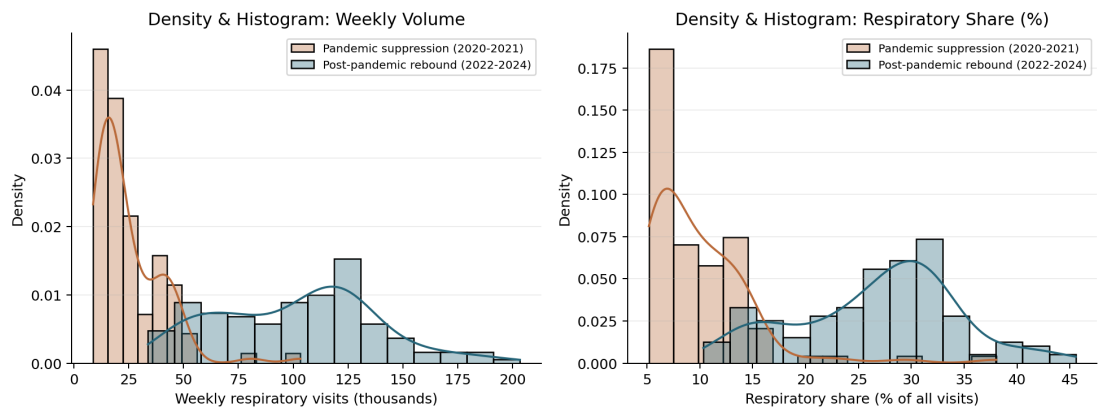


Figure 2b. Full EDA: Histograms and Kernel Density Estimation (KDE) comparing the pandemic suppression regime (2020-2021) with post-pandemic rebound (2022-2024).

Distributions and densities (Histograms & KDE): Weekly volume and share distributions reveal a marked regime shift. The pandemic era (2020-2021) is tightly concentrated at low demand (<50,000 weekly visits, median share ~10%) due to non-pharmacological interventions and school closures. In contrast, post-pandemic years (2022-2024) display a right-skewed, multimodal distribution with long seasonal tails extending to 203,353 visits and 45.56% share during winter peaks.

Year	Under 15	Age 65+	SAPU + SAR + SUR
2020	30.05%	12.31%	71.08%
2021	41.09%	9.47%	69.89%
2022	50.71%	7.33%	71.78%
2023	46.34%	8.81%	75.27%
2024	40.72%	9.81%	75.87%

Table 3. Shares of recorded respiratory visits, not population rates. The last column is place-of-care mix, not a measure of capacity or treatment success.

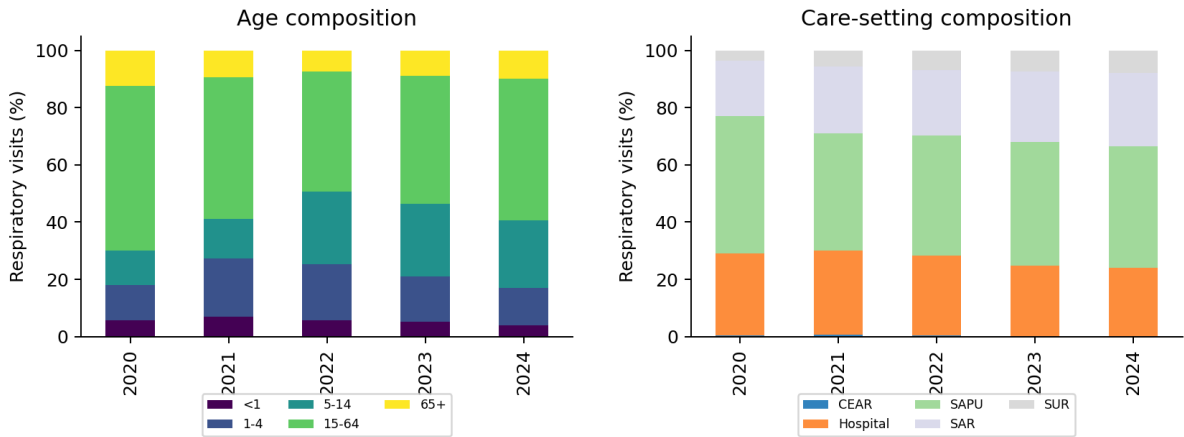


Figure 3. Age and facility-type composition of respiratory visits. Age groups reconcile to the respiratory Total. The place of care is not patient residence.

Under-15 visits represented 30.05% of respiratory visits in 2020 and 40.72% in 2024; SAPU, SAR and SUR together represented 71.08% and 75.87% respectively. This supports showing care-setting and age mix in a future monitoring view. It does not show whether hospital pressure was alleviated or whether children faced higher risk.

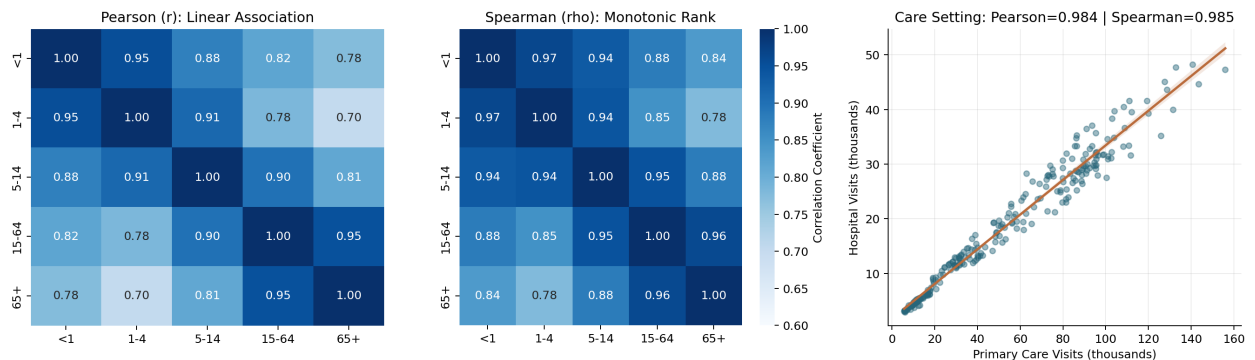


Figure 3b. Full EDA: Statistical evaluation of linear (Pearson r , left) vs monotonic rank (Spearman ρ , middle) co-variation across age bands, and care-setting coupling (right, $r = 0.984$, $\rho = 0.985$).

Parametric Pearson vs Nonparametric Spearman Analysis: Pearson correlation evaluates strictly linear co-variation under bivariate normality and is sensitive to heavy-tailed winter surges. Spearman rank correlation evaluates monotonic association without parametric distribution assumptions. Comparing both reveals that Spearman is systematically higher across age cohorts (Delta = +0.012 to +0.074). For example, between infants (<1) and seniors (65+), correlation increases from $r = 0.775$ (Pearson) to $\rho = 0.835$ (Spearman). This demonstrates that while epidemic wave slopes differ across cohorts (e.g. sharp early RSV surges vs delayed influenza peaks), their seasonal severity hierarchy is strongly monotonic. Meanwhile, primary care (SAPU/SAR/SUR) and hospital emergency departments move in near-lockstep across both metrics (Pearson $r = 0.9835$, Spearman $\rho = 0.9848$), confirming persistent structural operational coupling across all 260 complete weeks.

Year	All-facility share	Common-facility share	All-visit coverage
2020	10.75%	10.73%	98.05%
2021	9.97%	9.97%	97.31%
2022	24.69%	24.62%	96.44%
2023	28.95%	28.86%	96.36%
2024	28.51%	28.37%	95.89%

Table 4. A stable panel of 581 facility IDs reproduces the broad share pattern while covering at least 95.89% of all recorded visits in each year. This check addresses entry/exit, not every comparability concern.

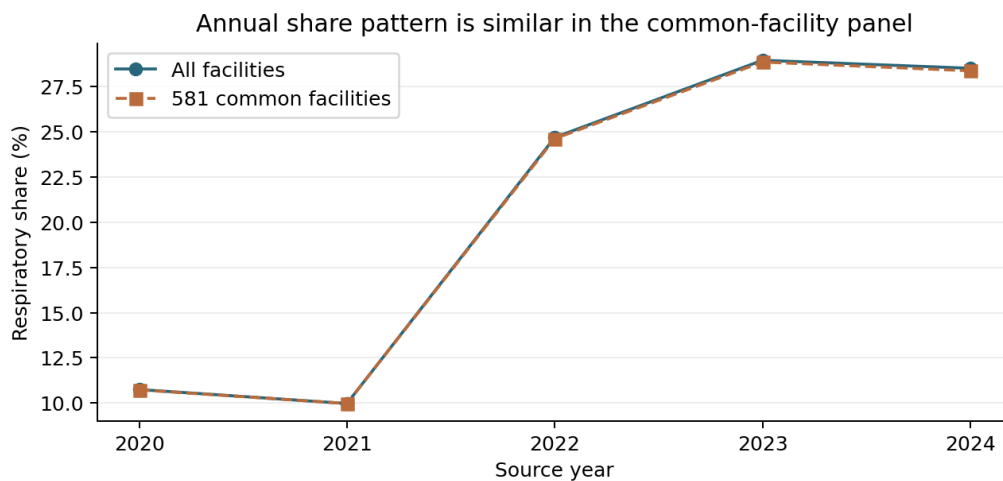


Figure 4. Full sample and 581 common-facility respiratory shares are close. Reporting completeness and pandemic-era conditions still limit interpretation.

6. Conclusion, limits and feasible continuation

The evidence answers the bounded question descriptively: recorded respiratory demand composition and complete-week peak timing varied markedly across 2020-2024; May peaks occurred in both 2023 and 2024; and the broad annual share change remains in facilities present across all years. For the proposed planning decision, a historical monitoring view should begin before midwinter and show current volume, share, age/care-setting mix and data coverage. Any operational shift or bed action requires current workload, staffing, occupancy, local catchment and clinical information not available here.

Alternative explanations include virus circulation, pandemic controls, care-seeking behavior, changes in reporting and the set of active facilities. ISP virological surveillance is a distinct, selected sample rather than a visit denominator; no causal or facility-level virological join is claimed. [5] There are no patient identifiers, residence or population denominator; counts cannot estimate incidence, unique patients, waiting times, outcomes, risk, care quality or the effect of an intervention. The 2020-2024 archives are historical and cannot themselves guide a live 2026 staffing decision.

Future ML and dashboard feasibility

A concrete future supervised task is a two-week-ahead forecast of completed-week respiratory visit counts by national care setting. One observation would be a week $\times$ care setting. Only information available before the forecast week - prior visit counts, calendar features and any verified contemporaneous external signal - could be inputs; the target week and post-outcome data cannot leak into predictors. Five annual cycles, the 2020-2021 regime change, facility reporting variation and lack of a current feed limit viability. The next stage should test whether a timely source and robust evaluation window exist before choosing a model. No ML is trained for C1.

A future dashboard for proposed winter planners would show weekly volume and share against comparable historical weeks, age distribution, hospital versus SAPU/SAR/SUR mix, and source freshness/coverage flags. It would support a human review of preparedness, not automatically prescribe capacity. A 30% alert threshold was a W1 design idea, not an official standard; it requires validation before use. No complete dashboard is built for C1.

7. Process record

Decisions and checks

Stage / decision	Alternative, reason and check	Evidence
W1 to C1 scope	Keep selected A; widen reduced input to every date of all five local years after internal team review. Check 2,169,824 selected rows and annual source coverage.	`data/input/`; `source_quality_profile.csv`; notebook sections 1-2
Duplicate handling	Retain raw rows in the extract; remove only two identical 2023 extras from outputs. Check full-row equality and 2023 change of -13 all visits, 0 respiratory.	Notebook section 2; `annual_summary.csv`
Week key	Do not group only by source year + `semana`: labels collide at calendar boundaries. Parse dates and use Sunday `week_start`; exclude two incomplete weeks from peak comparisons.	`source_week_collisions.csv`; notebook section 4
No geographic join	National question is answerable from shared fields. A facility-register join risks unmatched or duplicated geography without a validated historical mapping.	Notebook section 5; report sections 2-4
Coverage sensitivity	Compare all reporting facilities with 581 IDs present all five years; shares remain close, but reporting and pandemic limitations remain.	`facility_reporting_coverage.csv`; `common_facility_sensitivity.csv`
Reproducibility	Rebuild from the included uncleaned extract in a clean copied folder. Annual/weekly/profile outputs matched byte for byte after deterministic gzip metadata.	`analysis/prepare_reduced_case.py`; README

Instructor feedback and response

W1 was graded 100/100. The team reports no specific instructor comments after W1. Therefore there is no advice to claim as accepted, adapted or rejected. The wider-data correction was internal feedback from the teammate. The response is the complete five-year, all-date reduced case and auditable preparation described above. If the instructor later provides concrete guidance, the team should add its wording/date and response before submission.

Individual contributions

The previous submitted W1 report documents Benjamín Pinto as lead for Candidate A (respiratory emergency data, streamed processing, Five Cs, COVID sensitivity, multiyear figures and integrated comparison), and Sebastián Herrera as lead for Candidate B (REM-P2 codebooks, coverage anomalies, source inventory and nutrition discussion). This is the verified division of W1 work. The new C1 extract, notebook and report were prepared with Codex AI assistance under the team's request; the team must assign and verify each member's **actual C1 review, writing and presentation contribution** before submission. The file trail records work, but it cannot establish which student personally performed each new step. [7]

AI use and verification

Tool and prompt/output excerpt	Accepted, modified and verified
OpenAI Codex. Team request excerpt: 'Use all the years on the folder.' Output: scripts and a 2,169,824-row uncleaned ID 1/2 extract for 2020-2024.	Accepted scope after checking folder years, manifest hashes, row counts, nulls, zeros, duplicate rows and full-source rebuild.
OpenAI Codex. Team request excerpt: 'explain that the data has widened as a correction of the previous work.' Output: W1-to-C1 evolution statement.	Modified to distinguish W1's five-year aggregate analysis from its narrow submitted reduced input; checked against W1 PDF/notebook and C1 source files.
OpenAI Codex. Output: Sunday week key, common-facility sensitivity, executed notebook, figures and report draft.	Checked date/week collisions, annual-weekly reconciliation, cause pairing, identical duplicate contents, clean-folder rerun, saved notebook outputs and official source context. Team remains responsible for final interpretation.

Individual reflections

Benjamín Pinto - reflection carried forward from W1, to confirm for C1: contribution: efficient processing of the large emergency files and initial quality audit. Learning: administrative consultation counts differ from population prevalence. Remaining uncertainty: how to handle the 2020-2021 pandemic shock in a later forecasting model. **Sebastián Herrera - reflection carried forward from W1, to confirm for C1:** contribution: REM-P2 codebook and coverage audit that informed selection. Learning: a policy objective does not mean administrative data measure it without bias. Remaining uncertainty from W1: how to interpret absent REM-P2 code rows. Both members should revise these into their own current C1 reflections before submitting. [7]

References

- [1] Departamento de Estadísticas e Información de Salud (DEIS). Preguntas frecuentes: Atenciones de urgencia. Accessed 29 September 2026. deis.minsal.cl/faqs/.
- [2] Ministerio de Salud de Chile (MINSAL). Estrategia Nacional de Salud 2022. 2022, respiratory illness objectives and context. [MINSAL PDF](#).
- [3] MINSAL. Campaña Invierno 2024: informes de virus respiratorios. 2024. [MINSAL reports index](#).
- [4] MINSAL. Informe Campaña de Invierno, 24 May 2024, including week-20 emergency share. [MINSAL PDF](#).
- [5] Instituto de Salud Pública de Chile (ISP). 2024 respiratory-virus laboratory surveillance reports. [ISP index](#).
- [6] DEIS. Diccionario AtencionesUrgencia (local official workbook) and annual AtencionesUrgencia2020-2024 ZIP files; exact URLs, sizes, SHA-256 and retrieval date in `data/source\_manifest.csv`. Included reduced input and derivation scripts in this package.
- [7] Team. GROUP_02_W1_Report.pdf and W1 README, submitted English package, 25 September 2026, alternatives, Process record and full member names. The final C1 group ID must match W1.